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Measuring Diversity and Segregation with Convex Functions

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The study of quantitative measures for residential segregation extends far enough back that a review article in 1955 could express frustration and disappointment with the chaotic state of the methodology. Commenting then on the measures extant at the time, Duncan and Duncan [7] wrote:

1. *The proposed indexes of segregation have a number of hitherto unnoticed interrelationships which can be mathematically demonstrated.*
2. *Some of them have mathematical properties of which their proponents were unaware, and which lead to difficulties of interpretation.*
3. *As a consequence, the status of the empirical work already done with segregation indices is questionable, and their validity for further research is undermined.*

Over 70 years later, hints of this chaos remain. This author is aware of two instances in the last ten years of a segregation measure being published as novel despite already existing in the literature under a different name.

This article is not a systematic review of the literature, but rather a guided tour through one particular perspective on the problem of measuring segregation. This perspective begins with a general class of measures of diversity and segregation which

0.1.

Broad motivation for the article

- Folks in the math community are turning to quantitative measures of segregation [6, 8, 15].
- There's actually a pretty interesting and rich literature in quantitative sociology on this set of topics [12, 13], going back all the way at least to [14].
- MAUP is a thing [10], and we'd like measures that behave in predictable ways as we change analytical scale.

- If you start engaging with these ideas systematically and mathematically, then you come naturally to convexity and Bregman divergences, which are what I am here to sell you today.

Figure 1 is a simple checkerboard pattern. What if we actually oriented the entire piece around the checkerboard¹:

Preliminaries

To make these ideas concrete, we formalize the idea of a *population* and measurements which reflect the diversity or segregation of that population. Our framework for defining populations is intentionally discrete, in recognition of the fact that all Census-type data sets provide demographic information in discrete areal units such as counties, tracts, or blockgroups.

Definition 1. For $\ell \in \mathbb{N}$, the *probability simplex* Δ_ℓ is the set

$$\Delta_\ell = \left\{ \mathbf{y} \in \mathbb{R}^\ell : \sum_{j=1}^{\ell} y_j = 1, y_j \geq 0 \text{ for all } j \in [\ell] \right\}. \quad (1)$$

of probability vectors on ℓ elements.²

Definition 2. A *population* $\mathcal{P}$ with n locations in d spatial dimensions and ℓ demographic categories is a triple $\mathcal{P} = (\mathcal{X}, \mathbf{Y}, \mu)$, where:

- $\mathcal{X}$ is a finite metric space with n elements labeled $x_i, i = 1, \dots, n$.
- $\mathbf{Y} \in \mathbb{R}^{n \times \ell}$ is a row-stochastic matrix: for each row i , $\mathbf{y}_i \in \Delta_\ell$.
- $\mu \in \Delta_n$.

We denote the set of all populations with n locations, d dimensions, and ℓ demographic categories as $\mathcal{P}_{nd\ell}$.

In this definition, the matrix $\mathcal{X}$ represents space: element x_i of $\mathcal{X}$ can be viewed as giving the location of the i th spatial subunit of the population. The matrix $\mathbf{Y}$ represents the demographic distribution of the population according to some attribute such as gender, race, or education level: the i th row $\mathbf{y}_i$ of $\mathbf{Y}$ gives the proportion of individuals in each demographic category at location i . Finally, the probability vector μ represents the population density across locations; the proportion of individuals living in location i is μ_i . The probability that a single individual, chosen uniformly at random from the population, lives in location i and is in demographic category j is $\mu_i y_{ij}$. The mean population vector $\bar{\mathbf{y}} = \sum_{i=1}^n \mu_i \mathbf{y}_i$ has entries giving the marginal probabilities that an individual chosen uniformly at random belongs to each demographic category.

¹The problem of constructing segregation measures that distinguish between panels (c) and (d) in Figure 1 is often called the “checkerboard problem.”

²Although it is sometimes common to refer to Δ_ℓ as an $(\ell - 1)$ -simplex (this being its manifold dimension), we write Δ_ℓ to emphasize its role as the set of probability distributions over ℓ categories.

Definition 3. A *population measurement* is a function $f : \mathcal{P}_{nd\ell} \rightarrow \mathbb{R}$.³

An example of a population measurement is $f(\mathcal{P}) = \sum_{i=1}^n \mu_i^2$, which measures the degree to which the population is concentrated in space. For fixed n , the measure f is maximized (with value $f(\mathcal{P}) = 1$) when the entire population resides in a single location x_i and minimized (with value $f(\mathcal{P}) = \frac{1}{n}$) when the population is uniformly distributed across all locations.

Diversity and Convexity

Heuristically, we often say that a population is *diverse* with respect to an attribute when it contains many categories of that attribute in roughly equal proportions. Diversity is usually considered to be a property of a population as a whole, without regard for spatial or demographic subdivisions. In our formal structure from Definitions 2 and 3, a diversity measure $f : \mathcal{P}_{nd\ell} \rightarrow \mathbb{R}$ is a function of the mean population demographic vector: $f(\mathcal{P}) = \psi(\bar{\mathbf{y}})$ for some function $\psi : \Delta_\ell \rightarrow \mathbb{R}$. It remains to specify the properties that ψ should have to capture an intuitive notion of diversity.

Working in mathematical ecology, Pielou [11] gives a proposed set of criteria. For $\ell \in \mathbb{N}$, let $[\ell] = \{1, 2, \dots, \ell\}$. Let $\mathbf{e}_I \in \Delta_\ell$ be the vector with 1 in the indices specified by the multiindex $I \subseteq [\ell]$ and 0 elsewhere.

1. **Uniform maximizer.** “Diversity should be maximized when all groups are present in equal proportions.” In our setup, this means that $\frac{1}{\ell} \mathbf{e}_{[\ell]} = \operatorname{argmax}_{\mathbf{y} \in \Delta_\ell} \psi(\mathbf{y})$.
2. **Monotone under group addition.** “In two populations in which all groups are equally represented, the population with the greatest number of groups will be more diverse.” Mathematically, if I is a strict subset of $[\ell]$, then $\psi(\mathbf{e}_I / |I|) < \psi(\mathbf{e}_{[\ell]} / \ell)$.
3. **Additive under independent attributes.** “In a multiple classification (two or more qualitative variates) where the classifications are independent, the total diversity should equal the sum of the respective diversities.” This statement can be formalized as follows, interpreting “independent” in the sense of probability distributions: suppose that, for a given population $\mathcal{P} = (\mathcal{X}, \mathbf{Y}, \mu) \in \mathcal{P}_{nd\ell}$, there exist two other populations $\mathcal{P}_1 = (\mathcal{X}, \mathbf{U} \in \mathbb{R}^{n \times \ell_1}, \mu)$ and $\mathcal{P}_2 = (\mathcal{X}, \mathbf{V} \in \mathbb{R}^{n \times \ell_2}, \mu)$ such that $\ell = \ell_1 \ell_2$ and a bijection $\alpha : [\ell_1] \times [\ell_2] \rightarrow [\ell]$ such that $y_{ij} = u_{i\alpha_1(j)} v_{i\alpha_2(j)}$ for all $i \in [n]$ and $j \in [\ell]$. Then $f(\mathcal{P}) = f(\mathcal{P}_1) + f(\mathcal{P}_2)$.

We will soon find that properties (1) and (2) are satisfied by a broad range of functions ψ , while property (3) is more restrictive and essentially characterizes a single diversity measure.

Pielou’s proposed axioms capture some reasonable intuitions, but they don’t offer a recipe for constructing useful diversity functions ψ . A more constructive intuition is that *diversity should increase on average when populations are mixed*. If $\mathbf{y}$ and $\mathbf{y}'$ are two population demographic vectors and $\lambda \in (0, 1)$ is a mixing parameter, we can form a new demographic vector $\mathbf{y}'' =$

³The function η could reasonably be described as a *population measure*, which is an entirely disjoint concept from a population measurement. We will avoid the phrase “population measure” and emphasize “measurement” whenever possible.

$\lambda \mathbf{y} + (1 - \lambda) \mathbf{y}'$. One mathematization of our proposed intuition is that, if $\mathbf{y} \neq \mathbf{y}'$ then

$$\psi(\mathbf{y}'') > \lambda \psi(\mathbf{y}) + (1 - \lambda) \psi(\mathbf{y}') . \quad (2)$$

Here, the lefthand side gives the diversity of the mixed population $\mathbf{y}''$, while the righthand side gives the λ -weighted average diversity of the original populations $\mathbf{y}$ and $\mathbf{y}'$. This intuition leads us directly to the topic of convexity.

Definition 4. Set $S \in \mathbb{R}^d$ is a *convex set* if for all $x, y \in S$ and $\lambda \in [0, 1]$, we have $\lambda x + (1 - \lambda)y \in S$. If S is a convex set, then $f : S \rightarrow \mathbb{R}$ is a *convex function* if for all $x, y \in S$ and $\lambda \in [0, 1]$, we have $f(\lambda x + (1 - \lambda)y) \leq \lambda f(x) + (1 - \lambda)f(y)$. Function f is *strictly convex* if the above inequality is strict whenever $x \neq y$ and $\lambda \in (0, 1)$. Function f is (strictly) *concave* if $-f$ is (strictly) convex.

The simplex Δ_ℓ is a convex set, and so the notion of convexity applies to functions defined on Δ_ℓ . Equation (2) is precisely the condition that ψ is strictly concave. We will abuse notation slightly and refer to a population measurement $f(\mathcal{P}) = \psi(\bar{\mathbf{y}})$ as *concave* if ψ is concave.

Theorem 0.1. *If $\psi : \Delta_\ell \rightarrow \mathbb{R}$ is strictly concave, then the population measurement $f(\mathcal{P}) = \psi(\bar{\mathbf{y}})$ satisfies the monotonicity under group addition property.*

Proof. Monotonicity under group addition is a consequence of convexity and nonnegativity: if ψ is concave, then for any $I \subseteq [\ell]$ and $\lambda = \frac{|I|}{\ell}$, we have

$$\begin{aligned} \psi\left(\frac{1}{\ell} \mathbf{e}_{[\ell]}\right) &= \psi\left(\lambda \frac{1}{|I|} \mathbf{e}_I + (1 - \lambda) \frac{1}{\ell - |I|} \mathbf{e}_{[\ell] \setminus I}\right) \\ &\geq \lambda \psi\left(\frac{1}{|I|} \mathbf{e}_I\right) + (1 - \lambda) \psi\left(\frac{1}{\ell - |I|} \mathbf{e}_{[\ell] \setminus I}\right) \\ &\geq \psi\left(\frac{1}{|I|} \mathbf{e}_I\right) , \end{aligned}$$

where the first inequality is by convexity and the second is by nonnegativity. $\square$

Definition 5. Function f is *symmetric* if for all $\mathbf{y} \in \Delta_\ell$ and all permutations σ of $[\ell]$, we have $f(\mathbf{y}) = f(\sigma(\mathbf{y}))$, where $\sigma(\mathbf{y})$ is the vector obtained by permuting the entries of $\mathbf{y}$ according to σ .

Theorem 0.2. *If $\psi : \Delta_\ell \rightarrow \mathbb{R}$ is a nonnegative, symmetric, and strictly concave function, then the population measurement $f(\mathcal{P}) = \psi(\bar{\mathbf{y}})$ satisfies the uniform maximizer and monotonicity under group addition properties of Pielou's axioms.*

Proof. To demonstrate maximization under the uniform, for any vector $\mathbf{y} \in \Delta_\ell$,

$$\mathbf{e}_{[\ell]}/\ell = \frac{1}{|\mathcal{S}|} \sum_{\sigma \in \mathcal{S}} \sigma(\mathbf{y}) ,$$

where $\mathcal{S}$ is the set of all permutations of $[\ell]$. It follows that

$$\begin{aligned} \psi\left(\frac{1}{\ell} \mathbf{e}_{[\ell]}\right) &= \psi\left(\frac{1}{|\mathcal{S}|} \sum_{\sigma \in \mathcal{S}} \sigma(\mathbf{y})\right) \\ &< \frac{1}{|\mathcal{S}|} \sum_{\sigma \in \mathcal{S}} \psi(\sigma(\mathbf{y})) = \psi(\mathbf{y}) , \end{aligned}$$

where the inequality is by an inductive application of strict concavity and the final equality is by symmetry. $\square$

Definition 6. A function $\psi : \Delta_\ell \rightarrow \mathbb{R}$ is *separable* if there exists a function $\phi : [0, 1] \rightarrow \mathbb{R}$ such that $\psi(\mathbf{y}) = \sum_{j=1}^\ell \phi(y_j)$ for all $\mathbf{y} \in \Delta_\ell$.

Theorem 0.3. *If a function $\psi : \Delta_\ell \rightarrow \mathbb{R}$ is nonnegative, continuous⁴, symmetric, strictly concave, separable, and satisfies the additivity under independent attributes property, then⁵*

$$\psi(\mathbf{y}) = -c \sum_{j=1}^\ell y_j \log y_j, \quad (3)$$

for some constant $c > 0$.

A proof is given is given by Chakrabarti and Chakrabarty [3], where additivity under independent attributes corresponds to the authors' Axiom 4.

Equation (3) is the Shannon entropy, which is a widely used measure of diversity in ecology, sociology, and information theory. For a systematic treatment of entropy as a measure of diversity, see [9]. The function $\psi(\mathbf{y}) = 1 - \|\mathbf{y}\|_2^2 = 1 - \sum_{j=1}^\ell y_j^2$, the Simpson diversity index [14], is another popular measure of diversity which satisfies the uniform maximizer and monotonicity under group addition properties; it is separable and symmetric but does not satisfy additivity under independent attributes. The Simpson index has an attractive physical interpretation: it is the probability that two individuals chosen uniformly at random from the population belong to different demographic categories.

Segregation

Segregation is

PC: Need some notions from the sociology literature

One natural quantitative intuition is that a population is segregated when individual locations tend to be less diverse than the population as a whole.

Definition 7. Given a concave diversity function $\psi : \Delta_\ell \rightarrow \mathbb{R}$, the *Jensen information* [1] of a population $\mathcal{P} = (\mathcal{X}, \mathbf{Y}, \mu) \in \mathcal{P}_{nd\ell}$ is

$$I_\psi(\mathcal{P}) = \psi(\bar{\mathbf{y}}) - \sum_{i=1}^n \mu_i \psi(\mathbf{y}_i). \quad (4)$$

Jensen's Inequality, a standard result in convex analysis, implies that $I_\psi(\mathcal{P}) \geq 0$ for all populations $\mathcal{P}$ and concave functions ψ . If ψ is strictly concave, then $I_\psi(\mathcal{P}) = 0$ if and only if

⁴Concave functions are continuous on the interior of their domain, so this stipulation requires only that they are also continuous at the boundary.

⁵Throughout this article, we will use the notation $\log$ to denote the natural logarithm, as well as the standard convention that $0 \log 0 = 0$.

$\mathbf{y}_i = \bar{\mathbf{y}}$ for all $i \in [n]$, i.e. every location has the same demographic distribution as the population as a whole.

Figure 1 panels (a)-(c) give some intuition on how the diversity function ψ and the Jensen information I_ψ behave on a simple checkerboard pattern.

PC: place the measurements of entropy and mutual information in the titles for each panel.

Although the Jensen information eq. (4) is nonnegative, individual terms of the form $\psi(\bar{\mathbf{y}}) - \psi(\mathbf{y}_i)$ can be negative. This means that a given location's "contribution to segregation" cannot simply be measured by the difference $\psi(\bar{\mathbf{y}}) - \psi(\mathbf{y}_i)$. It would instead be convenient for analytic purposes to be able to write the Jensen information as a sum of nonnegative contributions from each location:

$$I_\psi(\mathcal{P}) = \sum_{i=1}^n \mu_i g(\mathbf{y}_i, \bar{\mathbf{y}}), \quad (5)$$

PC: Move from concave functions to functions whose negatives are convex.

Theorem 0.1 ([5]). *If $\phi : \Delta_\ell \rightarrow \mathbb{R}$ is a strictly convex function which is differentiable on the relative interior of Δ_ℓ and g is a function such that eq. (5) holds for any population $\mathcal{P} \in \mathcal{P}_{nd\ell}$, then g is the Bregman divergence induced by ϕ :*

$$g(\mathbf{y}, \bar{\mathbf{y}}) = D_\phi(\mathbf{y} \parallel \bar{\mathbf{y}}) = \phi(\mathbf{y}) - \phi(\bar{\mathbf{y}}) - \nabla \phi(\bar{\mathbf{y}})^\top (\mathbf{y} - \bar{\mathbf{y}}). \quad (6)$$

Literally, the Bregman divergence is a measure of the difference between the value of ϕ at $\mathbf{y}$ and the value of the linearization (first-order Taylor expansion) of ϕ around $\bar{\mathbf{y}}$ evaluated at $\mathbf{y}$. The Bregman divergence can be interpreted as a measure of dissimilarity of $\mathbf{y}$ from $\bar{\mathbf{y}}$. It is always nonnegative and is zero if and only if $\mathbf{y} = \bar{\mathbf{y}}$. The Bregman divergence is not a metric, as it is not symmetric and does not satisfy the triangle inequality, although it does satisfy many other useful properties [2]. Examples of Bregman divergences include the squared Euclidean distance $g(\mathbf{y}, \bar{\mathbf{y}}) = \|\mathbf{y} - \bar{\mathbf{y}}\|_2^2$ (corresponding to $\phi(x) = \frac{1}{2} \|x\|_2^2$) and the Kullback-Leibler divergence $g(\mathbf{y}, \bar{\mathbf{y}}) = \sum_{j=1}^\ell y_j \log(y_j / \bar{y}_j)$ (corresponding to $\phi(x) = \sum_{j=1}^\ell x_j \log x_j$).

PC: Locally quadratic, distance-like properties (foreshadowing for later)

0.1. Useful Properties of Bregman Divergences

- Additive decomposability by region.
- Migrations and exchanges
- Other?

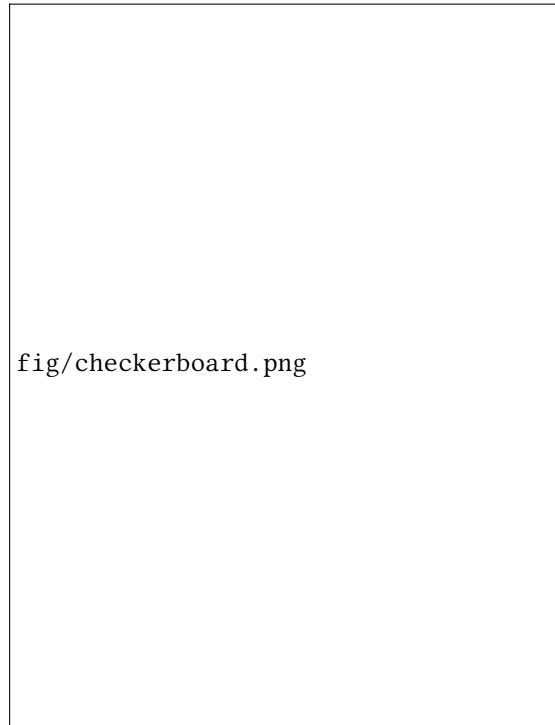


Figure 1: A simple checkerboard pattern. Each of the four panels is composed of 32 cells. Panel (c) is further arranged into 4 blocks of 8 cells each.

Spatializing the Jensen Information: Local Information

Extensions

- Side-by-side maps showing local information and maybe a regionalization.
- Possibly: scatterplot comparing local information and mutual information.

[5] [4]

[2]

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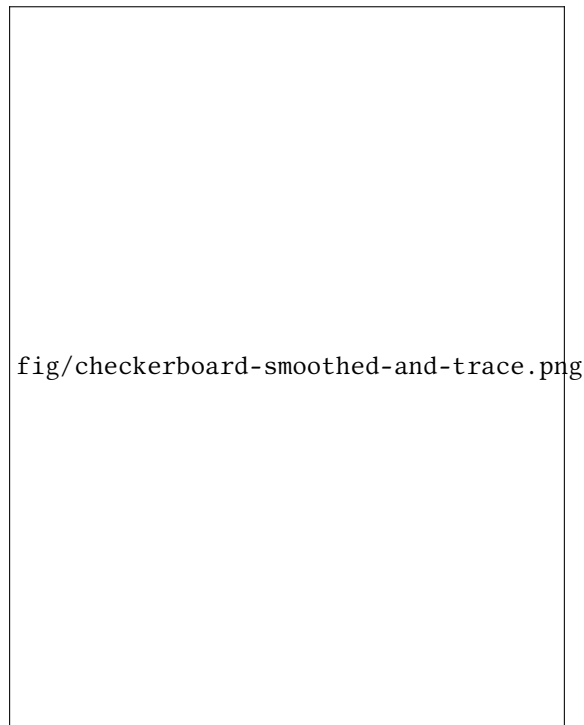


Figure 2: The smoothing bandwidth is 1 pixels. The mean local information of the checkerboard pattern is 0.228, while the mean local information of the segregated pattern is 0.054.

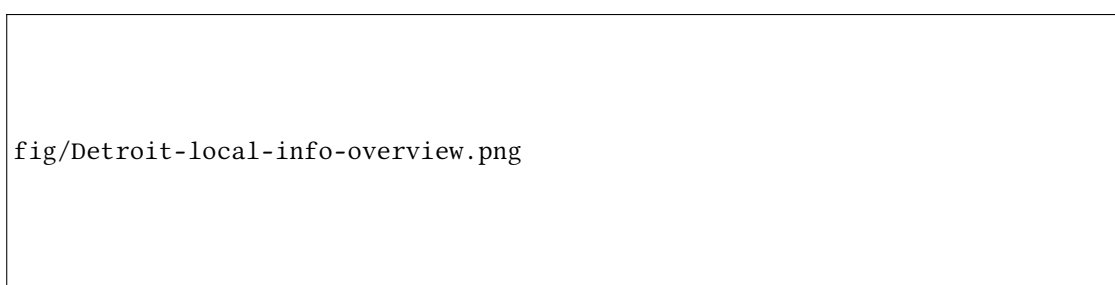


Figure 3: Test!!

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A. Supplementary Proofs

B. Supplementary Exercises

Exercise B.1. Give a heuristic justification for the convention that $0 \log 0 = 0$ by proving that $\lim_{x \rightarrow 0^+} x \log x = 0$.